

Machine Tool Technology

Week - 8

1. Introduction to Milling - Types of milling machine
- Specification of a Milling machine
2. Types of Milling cutters and their uses- Milling Cutter Nomenclature -Specification of Milling Cutter
3. Methods of Milling-Up Milling and Down Milling- Work holding devices

MILLING MACHINE

The milling machine is a type of machine which removes the material from the workpiece by feeding the work past a rotating multipoint cutter. The metal removal rate is higher very high as the cutter has a high speed and many cutting edges.

Milling machine is used for machining flat surfaces, contoured surfaces, surfaces of revolution, external and internal threads, and helical surfaces of various cross-sections.

Milling Machine Working Principle:

- Milling machine employed in the metal removing operation in which the work is rigidly clamped on the table of the machine and the revolving cutter which has multiple teeth is mounted on the arbor.
- The cutter revolves at high speed and the work is fed slowly past the cutter.
- The work can be fed in a vertical, longitudinal, or cross direction depending upon the type of milling machine being used.
- As the work proceeds, the cutter-teeth removes the metal from the surface of the job(workpiece) to produce the desired shape.

Types of milling machine

The milling machines are classified according to the general design of the machine.

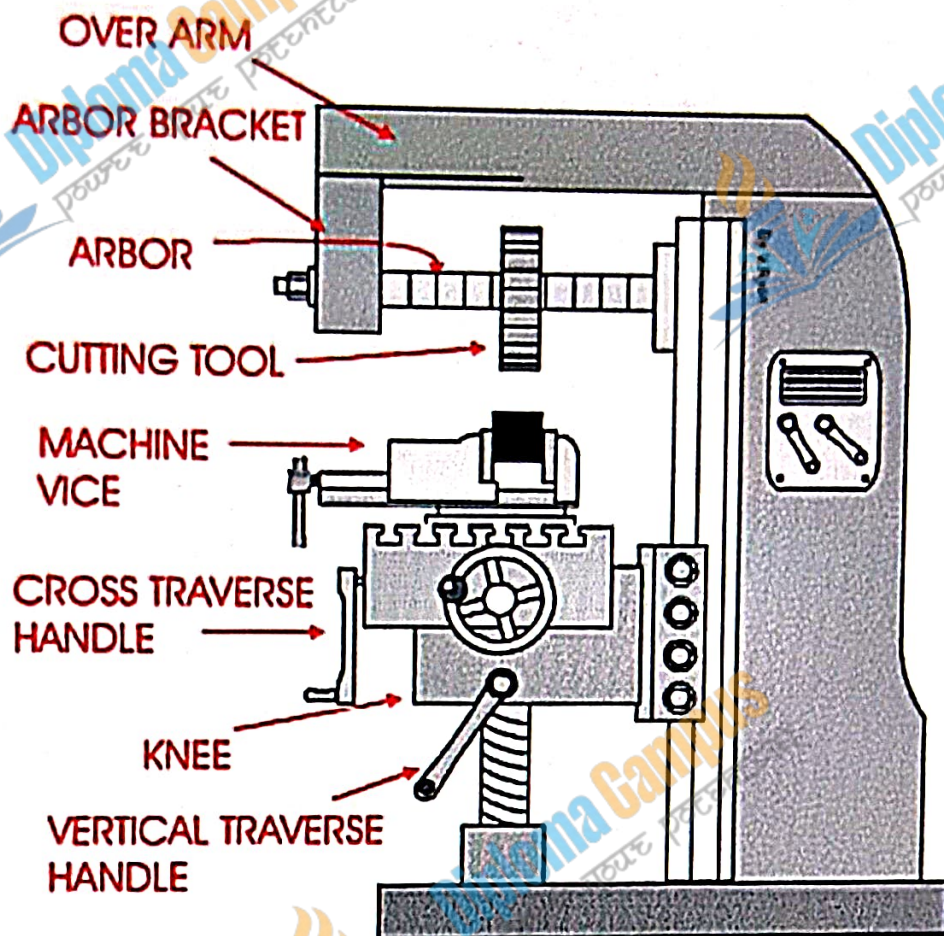
1. Column and knee type
 - a) Plain milling machine
 - b) Universal milling machine
 - c) Omniversal milling machine
 - d) Vertical milling machine
2. Table type milling machine
3. Planer type milling machine
4. Special type milling machine

Column and knee type milling machine

The column of a column and knee type milling machine is mounted vertically upon the base. Knee is mounted on the accurately machined guide ways of the column. It is designed to move up and down accurately. Saddle and table are mounted on the knee.

Main Parts of Column and knee type milling machine





Base

It is made of cast iron and supports all the other parts of the machine tool. A vertical column is mounted upon the base. In some machines, the base serves as a reservoir for cutting fluid.

Column

It is mounted upon the base and is box shaped. It houses the mechanism for providing drive for the spindle. The front vertical face of the column is machined accurately to form dovetail guide ways for the knee to move up and down. The top of the column holds an overhanging arm.

Knee

It slides up and down on the guide ways of the column. An elevating screw mounted on the base obtains this movement. Saddle is mounted upon the knee and moves in a cross direction.

Saddle

It is mounted on the guide ways of the knee and moves towards or away from the face of the column. This movement can be obtained either by power or by hand. The top of the saddle has guide ways for the table movement.

Table

The table is moved longitudinally either by power or manually on the guide ways of the saddle. The trip dogs placed on it control the movement of the table. The table of a universal milling machine can be swivelled horizontally to perform helical works. The top surface of the table has got 'T' – slots on which the workpieces or other work holding devices are mounted.

Spindle



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It is located in the upper part of the column. It receives power from the motor through belt, gears and clutches. The front end of the spindle has got a taper hole into which the cutters are held with different cutter holding devices.

Overhanging arm

It supports the arbour from the top of the column. The arbour is supported by the bearing fitted within the arbour support. It is also useful while using some special attachments.

Front brace

It is an extra support fitted between the knee and the overhanging arm. It is slotted to allow the knee to be adjusted vertically.

Arbor

It supports the different types of cutters used in the machine. It is drawn into the taper hole of the spindle by a draw bolt. One or more cutters are mounted on the arbour by placing spacing collars

between them. The arbour is supported by an arbour support. The arbour is provided with a Morse taper or self-releasing taper.

There are different types of column and knee type machines.

a) Plain milling machine or Horizontal milling machine

It is rigid and sturdy. Heavy workpieces are mounted and machined on the machine. The work mounted on the table is moved vertically, longitudinally and crosswise against the rotating cutter. The table cannot be rotated. It is also called as horizontal milling machine because the cutter rotates in horizontal plane.

b) Universal milling machine

The table of a universal milling machine can be swivelled by 45° on either side and so helical milling works can be performed. It is named so because it can be adapted for a very wide range of milling operations. Various milling attachments like index head, vertical milling head, slot milling head and rotary table can be mounted. It can machine drills, reamers, gears, milling cutters with a very high degree of accuracy and so it finds an important place in a workshop.

c) Omniversal milling machine

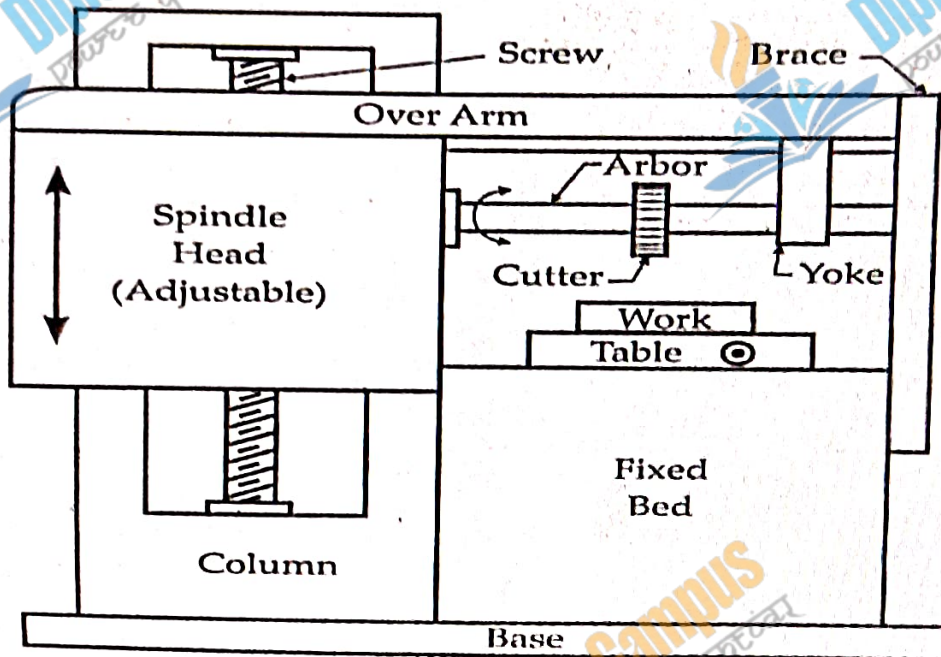
In addition to the table movements obtained in a universal milling machine, the knee can be tilted to a required angle. It is useful for machining helical grooves, reamer and bevel gears. It is mostly used in tool room work.

d) Vertical milling machine

It is very similar to a horizontal milling machine in construction as it has the same parts of base, column, knee, saddle and table. The spindle of the machine is positioned vertically. The cutters are mounted on the spindle. The spindle is rotated by the power obtained from the mechanism placed inside the column. Angular surfaces are machined by swivelling the spindle head.

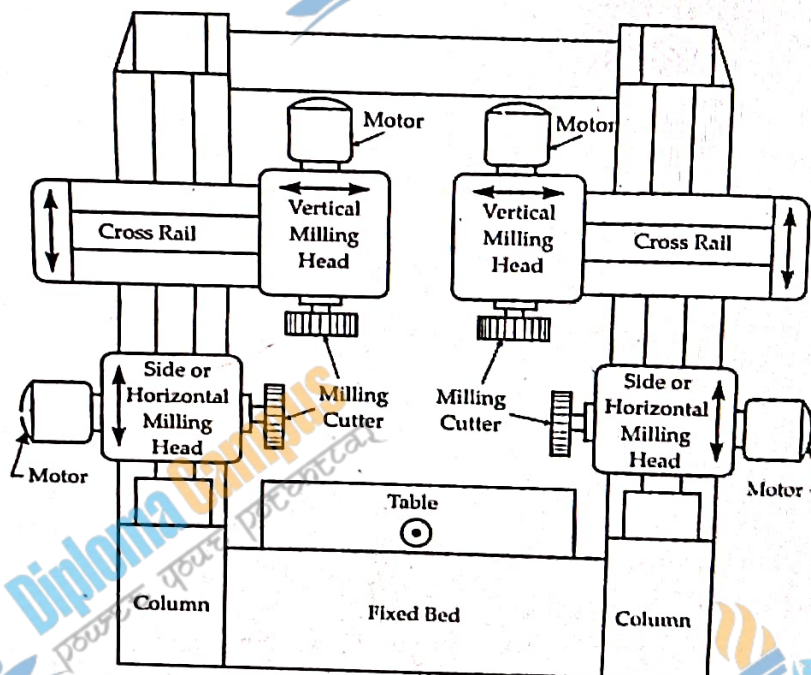


2. Fixed bed milling machine



In this type of machine, the bed of the machine is fixed to the machine. No arrangement of the knee and saddle can move vertically and transversally. The worktable is directly situated on the fixed bed. The spindle of this machine is mounted on a movable spindle head. It can move in the vertical and horizontal direction and perform the cutting operation.

3. Planer milling machine



The planer milling machine is mostly used for facing operations in mass production. These machines are similar to the bed-type mill machine, except they can be mounted with various

cutters and spindle heads to the machine. These cutters in the machines can perform the facing operations simultaneously which is a great function.

4. Special milling machine

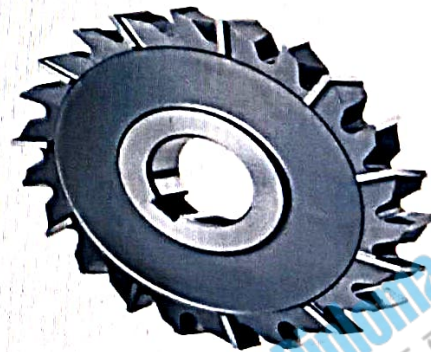
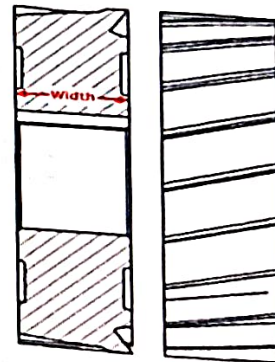
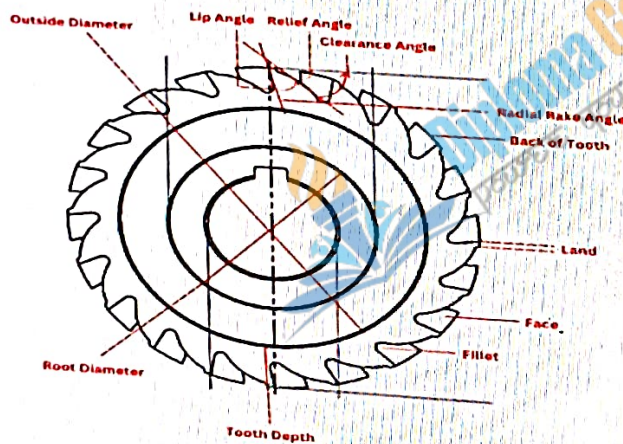
These machines are modern milling machines that are developed to ease milling operations according to the jobs.

Size of a milling machine

The size of a milling machine is specified as follows

1. The size of the table (length and width)
2. The maximum lengths of longitudinal, cross and vertical travel of the table.
3. Number of spindle speeds, number of feeds
4. Spindle nose taper
5. Power required
6. Net weight of the machine
7. The floor space required
8. Type of the machine

Milling Cutter Nomenclature-Specification of Milling Cutter



Body of cutter: It is the main frame of milling cutter, on which the teeth rest.

Periphery: It is defined as the locus of cutting edges of tooth of cutter.



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Cutting edge: It is the portion that touches the workpiece during cutting action. It is the intersection of teeth face and tooth flank.

Fillet: portion where one teeth joins the face of another tooth. It is a reinforcement to cutting tooth.

Face of teeth: it is the surface upon the chip is formed while cutting. It may be curved or flat.

Back of tooth: it is the created by fillet and the secondary clearance angle.

Land: it is the narrow surface on the back of cutting edge. Land is the result of providing the clearance angle.

Bottom Land: the blank space between the consecutive teeth.

Root diameter: diameter passing through centre of cutter and joining two ends of the periphery.

Root diameter: passing through centre of cutter and joining two bottom fillet.

Angles of milling cutter

Lip angle: It is the angle between the face of the teeth and the land of the teeth. Land is a narrow surface on the back of teeth.

Relief angle: it is the angle between the tangent to the outside diameter of the cutter at cutting edge and the land of the tooth. The function relief angle is to avoid the interference between the land of the tooth and the work surface. The relief angle varies with the type of material to be machined.

Radial rake angle: it is the angle between the face of teeth and the radial line passing through the cutting edge of the tooth. The radial rake angle may be positive, negative or zero. It is provided free cutting by allowing chips to flow smoothly on the face of the cutter.

Zero rake angle: tooth face and radial line coincide.

Positive rake angle: tooth body and tooth face is the same side of the radial line. **Negative rake angle:** tooth surface and radial angle are on opposite side of the radial line.

Axial rake angle: angle between the line of peripheral cutting edge and axis of the cutter. It is an angle in a plane perpendicular to the radial plane.

Clearance angle: it is the angle between the back of the tooth and the tangent to the outside diameter at the cutting edge of the tooth. It is divided into two according to the clearance surface. (primary clearance angle and secondary clearance angle)

Milling cutters

Milling cutters are crucial components of milling machines, serving as the cutting tools used to remove excess material from the workpiece. Each milling machine incorporates a specific cutter for precise machining operations.

Types of Milling Cutting Tools



The various types of milling cutters are:

1. Plain Milling Cutters
2. Face Milling Cutter
3. Side milling cutter
4. Angle Milling Cutters
5. 'T' Slot Milling Cutter
6. Form Relieving Cutter
7. Gang Milling Cutter
8. Peripheral Milling cutter
9. Dovetail cutter
10. End Mill Cutter
11. Fly Cutter
12. Woodruff Key Slot Milling Cutter
13. Thread mill Cutter
14. Ball cutter
15. Hobbing cutter

Plain Milling Cutters:

Plain milling cutters possess a cylindrical shape and feature teeth solely on the periphery surface. These versatile milling cutters are specifically employed for creating flat, plane, and horizontal surfaces in various machining operations.

Side Milling Cutter:

A side milling cutter is a type of cutting tool with teeth on its side circumference. It is designed for removing material from the side of a workpiece.

Side milling cutters are used in various machining operations, including slotting, contouring, and angular surface machining, making them versatile tools for different applications.

Angle Milling Cutters:

Angle milling cutters are specialised cutting tools with angled teeth designed to create chamfers, bevels, and angles on workpiece edges, enhancing their visual appeal and functionality.

T-Slot Milling Cutter:

T-Slot Milling Cutter is a specialised tool designed for machining T-slots in workpieces. It features a unique design with a T-shaped groove that matches the T-slot profile.

T-Slot cutters come in various sizes to accommodate different T-slot dimensions.

They are commonly used in the manufacturing of fixtures, jigs, and work holding devices.

Form Relieving Cutter:

A form relieving cutter is a specialised tool used to create intricate profiles on workpieces.

Due to their precision and ability to follow intricate profiles, form relieving cutters often produce superior surface finishes, minimising the need for additional finishing processes.

Gang Milling Cutter:

Gang milling cutters consist of two or more individual cutters mounted on the same arbor, allowing simultaneous cutting of multiple surfaces or features on a workpiece.

This setup enables increased productivity as multiple cuts are performed in a single pass, reducing overall machining time and increasing efficiency.





Gang milling cutters are often used for machining complex profiles, contours, and slots where multiple cutting actions are required on the same workpiece.

Peripheral Milling cutter

A peripheral milling cutter is a type of cutting tool used for removing material from the external surface of a workpiece. It operates by cutting along the periphery of the cutter's circumference. Peripheral milling cutters are versatile and can be used for various machining operations, including facing, contouring, slotting, and producing grooves on the outer surface of the workpiece.

End Mill Cutter:

An end mill cutter is a versatile rotary cutting tool with multiple flutes that can be used for various milling operations, including contouring, slotting, and pocketing.

Fly Cutter:

A fly cutter is a single-point cutting tool used in machining operations, featuring a large, flat cutting surface that rotates to remove material from the workpiece.

Woodruff Key Slot Milling Cutter:

A Woodruff Key Slot Milling Cutter is a specialised tool designed to create precise keyways or slots in shafts, hubs, and other components to accommodate Woodruff keys, ensuring secure mechanical connections.

Dovetail cutter

A dovetail cutter is characterised by its unique triangular profile, resembling the shape of a dovetail joint. This profile is ideal for creating dovetail grooves or slots in workpieces.

Thread mill Cutter

A thread mill cutter is a specialised tool used for machining threads in workpieces, creating internal or external threads with precision and accuracy.

Hobbing cutter

A hobbing cutter is a tool used for manufacturing gears. It employs a cutting action to generate gear teeth with accuracy and consistency.





Roughing End Mill



Slab Mill Cutter



End Mill Cutter



Ball Mill Cutter



Hollow Mill Cutter



Wood Ruff Cutter



Thread Mill Cutter



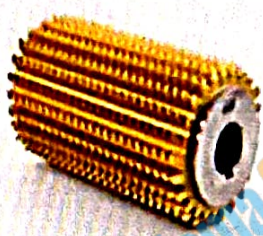
Fly Cutter



Face Mill Cutter



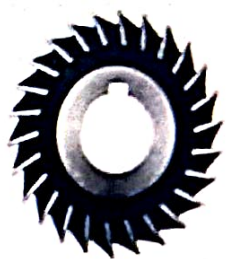
Involute Gear Cutter



Hobbing Cutter



Dovetail Cutter



Slide and Face Cutter

Milling Operation - Peripheral milling

The machining is performed by the cutting edges on the periphery of the milling cutter.

It is classified under two

1. Up milling
2. Down milling

Up milling

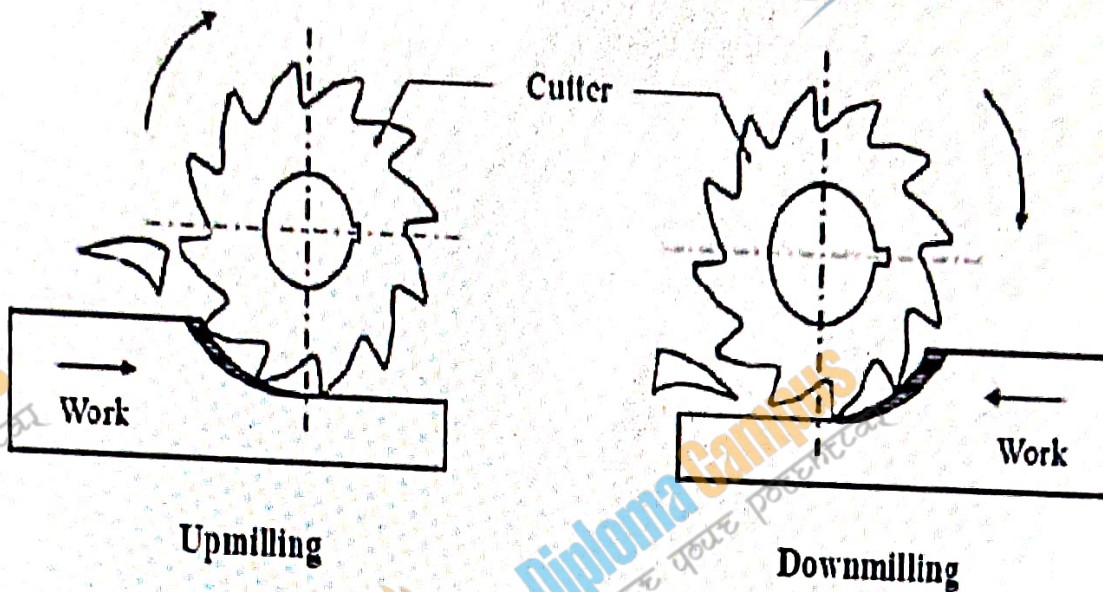
In this method, the workpiece mounted on the table is fed against the direction of rotation of the milling cutter. The cutting force is minimum during the beginning of the cut and maximum at the end of cut. The thickness of the chip is more at the end of the cut. As the cutting force is directed upwards, it tends to lift the workpiece from the fixtures. A difficulty is felt in pouring coolant on the cutting edge. Due to these reasons the quality of the surface obtained by this method is wavy. This process being safer is commonly used and sometimes called conventional milling.

Down milling

The workpiece mounted on the table is moved in the same direction as that of the rotation of the milling cutter. The cutting force is maximum at the beginning and minimum at the end of cut. The chip thickness is more at the beginning of the cut. The workpiece is not disturbed because of the bite of the cutter on the work. The coolant directly reaches to the cutting point. So the



quality of surface finish obtained is high. Because of the backlash error between the feed screw of the table and the nut, vibration is setup on the workpiece.



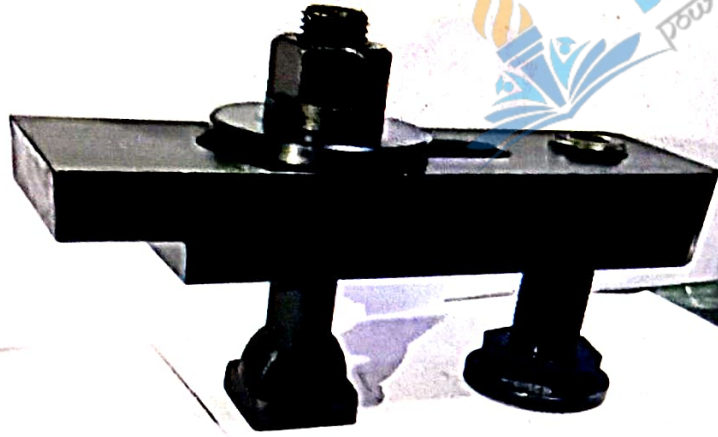
Work holding devices used for milling machine

Various types of work holding devices are used for milling machine operations they are explained as follows :

1. T-bolts and clamps
2. Angle Plates
3. V – Block
4. Machine Vices

1.T- Bolts and clamps –

Bulky workpieces of irregular shapes are clamped directly on the milling machine table by using T- bolts and clamps. Different types of clamps are used for different patterns of work. All these clamps carry a long hole, through which clamping bolt passes. This hole permits the bolts for adjustment according to the size and shape of the job.



2. Angle plates –

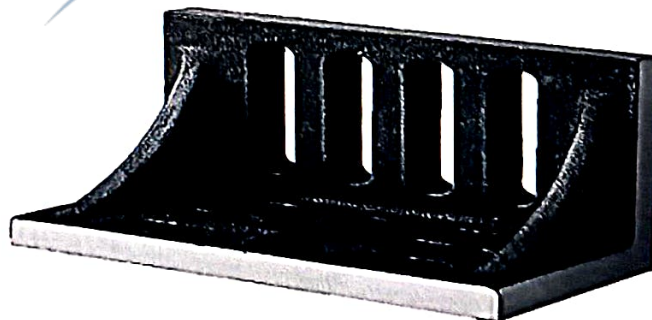
When work surfaces are to be milled at right angles to another face, angle plates are used for supporting the work.

The angle plate is made from high-quality material (generally spheroidal cast iron) that has been stabilized to prevent further movement or distortion.

Slotted holes or "T" bolt slots are machined into the surfaces to enable the secure attachment or clamping of workpieces to the plate, and also of the plate to the worktable.

Angle plates also may be used to hold the workpiece square to the table during marking-out operations.

Adjustable angle plates are also available for workpieces that need to be inclined, usually towards a milling cutter.



3. V block –

The V blocks are used for holding shafts on a milling machine table in which keyways and slots are to be milled.



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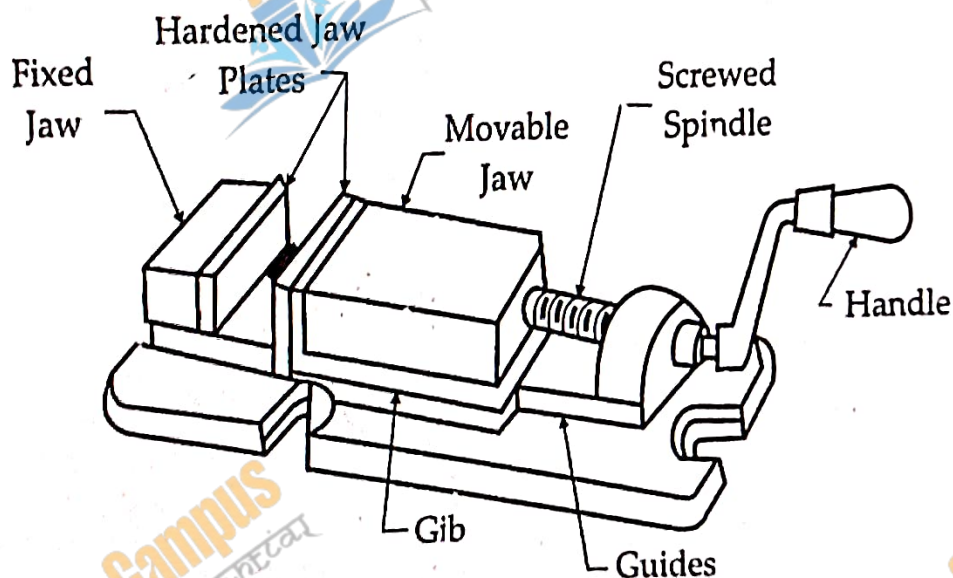


4. Vices –

Vices are the most common appliance for holding work on milling machine tables. According to its quick loading and unloading arrangement. Vices are of three types,

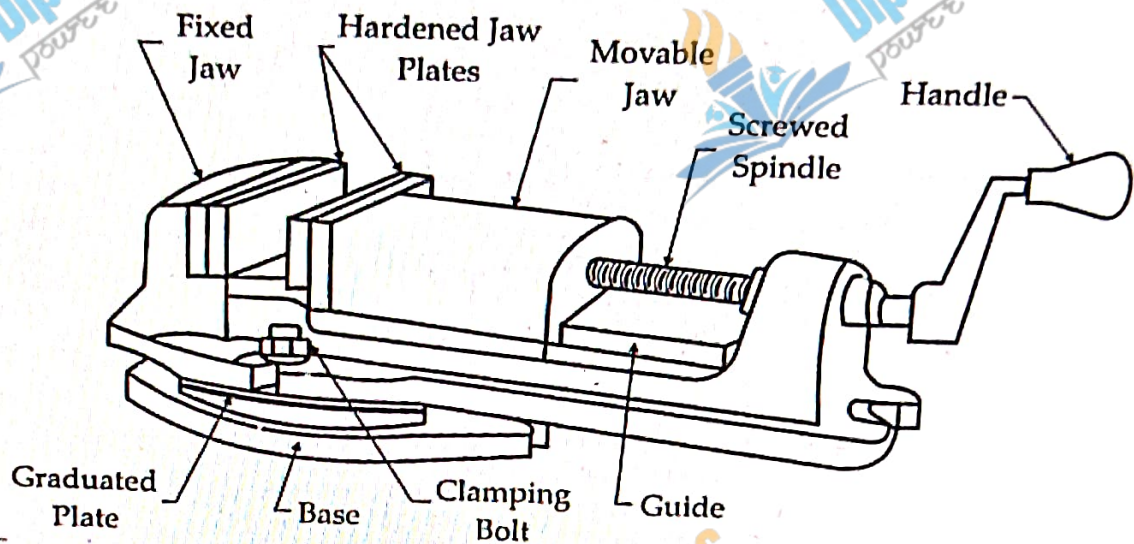
(a) Plain Vice –

The plain vice is directly bolted on the milling machine table. It is the most common type of vice used on plain milling operations, which involves heavy cuts, such as in slab milling. Its especially low construction enables the work to remain quite close to the table. This reduces the chance of vibration to a minimum. The base carries slots to accommodate 'T' bolts to fix the vice on the table. Work is clamped between the fixed and movable jaw and for holding workpieces of irregular shape special jaws are sometimes used.



(b) Swivel Vices –

The swivel vice is used to mill an angular surface in relation to a straight surface without removing the work from the vice. It has got a circular base graduated in degrees. The base is clamped on the table by means of T- bolts.



(c) Universal Vices –

It can be swiveled in a horizontal plane similar to a swivel vice and can also be tilted in any vertical position for an angular cut. The vice is not rigid in construction and is used mainly in tool room work. It enables the milling of various surfaces, at an inclination to one another, without removing the workpiece.

